

Earth

Overview

Earth is the third planet from the Sun and the only astronomical object known to harbour life, making it uniquely precious in the known universe. It orbits the Sun at an average distance of about 150 million km — a distance defined as one astronomical unit (AU) — completing one orbit every 365.25 days. Earth's diameter of 12,742 km makes it the largest of the four terrestrial planets. Its position in the Sun's habitable zone — sometimes called the Goldilocks zone — means liquid water can exist stably on its surface, a prerequisite for life as we know it. Earth has a mean surface temperature of about 15 °C, maintained by the natural greenhouse effect of its atmosphere, which adds roughly 33 °C of warming to what would otherwise be a frozen world.

Atmosphere

Earth's atmosphere is composed primarily of 78 % nitrogen (N₂) and 21 % oxygen (O₂), with argon at 0.93 % and carbon dioxide at about 0.04 %, plus trace amounts of other gases including water vapour, which varies from 0–4 % and plays a crucial role in weather and climate. The ozone layer, concentrated in the stratosphere at 15–35 km altitude, absorbs most of the Sun's harmful ultraviolet-B and ultraviolet-C radiation, protecting life on the surface from DNA-damaging rays. Earth's magnetic field, generated by convection in its liquid iron outer core, deflects the solar wind and cosmic rays, preventing atmospheric erosion and shielding the surface from harmful radiation. Without this magnetosphere, Earth might have lost its oceans and atmosphere to space long ago, as Mars appears to have done. Human industrial activity has raised atmospheric CO₂ from about 280 ppm before the Industrial Revolution to over 420 ppm today, intensifying the greenhouse effect and driving observable climate change.

Hydrosphere and Oceans

Water covers approximately 71 % of Earth's surface, with the world's oceans holding about 97 % of all surface water and averaging 3,688 metres in depth. The Pacific Ocean alone is larger than all of Earth's landmasses combined, covering about 165 million km², and the Mariana Trench within it plunges to a depth of 11,034 metres — the deepest known point on Earth. Ocean currents act as a global conveyor belt, redistributing heat around the planet and profoundly influencing regional climates; the Gulf Stream, for example, keeps Western Europe much warmer than it would otherwise be at those latitudes. Earth's water cycle — evaporation, condensation, precipitation, and runoff — continuously moves water between the oceans, atmosphere, and land, making fresh water available for terrestrial life. About 1.7 % of all water is stored in glaciers and ice caps, which act as long-term reservoirs; their retreat due to climate change is raising sea levels at a rate of approximately 3.3 mm per year.

Geology and Plate Tectonics

Earth is the only planet in the Solar System with confirmed active plate tectonics, a process in which the rigid lithosphere is broken into about 15 major and several minor tectonic plates that drift over the semi-fluid asthenosphere below. These plates move at speeds of 2–15 cm per year — roughly the rate at which fingernails grow — driven by convection currents in the mantle and the sinking of dense oceanic crust at subduction zones. Where plates collide, mountain ranges are thrust upward; the Himalayas, for example, are growing because the Indian Plate is colliding with the Eurasian Plate. Where plates separate, new oceanic crust is created at mid-ocean ridges such as the Mid-Atlantic Ridge, which is spreading at about 2.5 cm per year. Plate tectonics is also the primary driver of Earth's long-term carbon cycle: volcanic outgassing releases CO₂, while subduction of carbonate rocks returns carbon to the mantle, regulating atmospheric CO₂ over millions of years and keeping Earth's climate relatively stable.

The Moon

The Moon is Earth's only natural satellite and the fifth largest moon in the Solar System, with a diameter of 3,474 km. The leading hypothesis for its origin is the Giant Impact theory: approximately 4.5 billion years ago, a Mars-sized protoplanet called Theia collided with the proto-Earth at a glancing angle, ejecting a vast cloud of debris that coalesced into the Moon over millions of years. The Moon stabilises Earth's axial tilt at about 23.5°, which moderates seasonal variation and has been important for maintaining a stable climate conducive to life. Tidal forces between Earth and the Moon are gradually slowing Earth's rotation and causing the Moon to recede from Earth at about 3.8 cm per year. The Moon's gravity is the primary driver of Earth's ocean tides, and tidal locking means the Moon always shows the same face toward Earth. The Apollo missions (1969–1972) brought back 382 kg of lunar rock and soil, providing the most detailed direct knowledge of any extraterrestrial body.